Reproducing MG-GAT: Interpretable Rating Prediction with Geometric Deep Learning

Paper: Leng, Liu & Ruiz — *Interpretable Recommendations and User-Centric Explanations with Geometric Deep Learning* (SSRN 3696092)
Task: Stage 1 only (rating prediction, no LLM). **Target**: Test RMSE ≈ 1.249

1. Problem & Scope

MG-GAT is a Multi-Graph Graph Attention Network for **matrix completion**: predicting how a user would rate a business they have not visited. The user-business rating matrix X is extremely sparse (fill rate < 0.012%). The model uses side features (S_u, S_b) and graph structures (G_u, G_b) to overcome this sparsity.

Per assignment requirements, all LLM components were excluded: the 4th business graph (LLM perceptual map), Algorithm 2 (explanation generation), and Section 4 of the paper. Only 3 business graph types were retained.

2. Data

2.1 Dataset Version & Scale

The original paper used the 2019 Yelp dataset (no longer publicly available). This work uses the 2024 version, which is significantly larger but sparser.

Statistic	This Work	Paper (Table D1)	Ratio
Users	320,212	76,865	4.2×
Businesses	31,663	10,966	2.9×
Total reviews	1,183,126	260,350	4.5×
Train reviews	820,496	~180,000	4.6×
Val reviews	179,662	~40,000	4.5×
Test reviews	182,968	~40,000	4.6×
Avg reviews/user	3.695	3.387	similar ✓
Avg reviews/biz	37.37	23.742	1.6×
Avg user friend degree	5.449	5.557	similar ✓
Biz graph avg degree	15.6	30.0	0.52×
Mean stars	3.698	3.728	similar ✓
Std stars	1.428	1.384	similar ✓

2.2 Time Split

Split	Period	Reviews
Train	2009–2016	820,496
Validation	2017	179,662
Test	2018	182,968

Pipeline: filter PA businesses from `business.json` → filter reviews by PA business IDs and year range → extract user IDs → extract features from `user.json` + `checkin.json` → split by year.

2.3 Features

User features S_u (65 dims = 33 explicit + 32 implicit SVD): compliments (11), votes (3), fans (1), yelping_since year/month/day (3), elite 2005–2018 + None (15). All explicit features min-max normalised.

Business features S_b (1,428 dims = 1,396 explicit + 32 implicit SVD): attributes 87 one-hot cols (paper: 93); categories 1,149 one-hot multi-label cols (paper: 946) — differences due to dataset version; hours 14 cols (open/close × 7 days); check-in 144 cols (7 days × 24 hourly bins); location 2 cols (lat/lon).

Implicit features: binarise training ratings → truncated SVD (k_i=32) → concatenate with explicit features. SVD uses **training data only** to prevent leakage:

```
X_bina = U(0) · Σ · B(0)
```

```
Su,imp = U(0)  · Σ^(1/2)     shape: (n × 32)
Sb,imp = B(0)T  · Σ^(1/2)    shape: (m × 32)
```

2.4 Graphs

Graph	Construction	k	Avg deg (ours)	Avg deg (paper)
G_u (friendship)	Yelp friends field, undirected	—	5.449	5.557
G_b geo	Haversine distance, k-NN, OR symmetrisation	10	~12.7	~10
G_b co-visit	Shared customers (train only), X^TX, OR sym.	10	~16.2	~10
G_b category	Shared category count, MM^T, OR sym.	10	~18.5	~10

Three graphs are stored independently and combined via learnable ω_g weights — they are **not** merged into a single adjacency matrix.

3. Model Architecture

3.1 Five-Layer Network

Layer	Equation	Purpose	Key Detail
1 — Linear	$H(1) = W(1) \cdot S$	Project raw features	No bias, no activation — preserves FR interpretability
2 — Attention (NIG)	$\alpha = \text{softmax}(\text{LReLU}(a^T \cdot [H(1)_i \parallel H(1)_k]))$	Compute neighbor importance	Biz side: 3 graphs weighted by ωg, unified softmax
3 — Aggregation	$H(2)_i = \Sigma k \in Ni\ \alpha \cdot H(1)_k$	Aggregate neighbor info	Neighbours only, excludes self
4 — Nonlinear	$H(3) = \text{actv1}(W(2) \cdot H(2) + W(2)s \cdot S + b)$	Nonlinear transform + skip	Skip connection from raw features S
5 — Final	$U_i = \text{actv2}(W(3) \cdot H(3)_i) + H(4)_i$	Final embedding	H(4) = free embedding, graph-Laplacian regularised

Business-side multi-graph NIG:

```
α(l→j)_b = softmax_j( LeakyReLU( Σ_{g∈B} ω_g · a_bᵀ · [H(1)_j ‖ H(1)_l] ) )
```

3.2 Prediction & Loss

Rating prediction: $\hat{X}_{ij} = (r_{max} - r_{min}) \cdot \sigma(U_i \cdot B_j^T + b_{u,i} + b_{b,j} + b) + r_{min} = 4 \cdot \sigma(\ldots) + 1$

Loss: $L = \|\Omega_{train} \circ (X - \hat{X})\|^2 + \theta_1 \cdot [\text{Tr}(H^{(4)T}_u \tilde{L}_u H^{(4)}_u) + \text{Tr}(H^{(4)T}_b \tilde{L}_b H^{(4)}_b)]$ where $\tilde{L} = L + \theta_2 \cdot I$

Key constraints: Graph regularisation applies **only to $H^{(4)}$, not $H^{(3)}$** — because Layers 2–3 already enforce local smoothness via attention. Loss computed only at observed positions. All parameters have L2 regularisation. Optimiser: Adam + ReduceLROnPlateau.

3.3 Interpretability

Feature Relevance (FR): $FR = a^T \cdot W^{(1)}$ — directly readable because Layer 1 is linear (no bias, no activation). Reveals which input features drive predictions globally.

```
FR_user_self = a(u,self)ᵀ·W(1)_u     FR_user_nb = a(u,nb)ᵀ·W(1)_u
FR_biz_self  = a(b,self)ᵀ·W(1)_b     FR_biz_nb  = a(b,nb)ᵀ·W(1)_b
```

Neighbor Importance Graph (NIG): attention weights α(k→i) represent neighbor importance; ω_g show relative graph-type importance.

4. Hyperparameter Search

The paper uses Hyperopt (TPE) but does not disclose final values. We ran **50 trials × 30 epochs** across the following search space:

Parameter	Search Space	Parameter	Best Value
θ_1	{0.001, 0.01, 0.1}	θ_1	0.01
θ_2	{0.001, 0.01, 0.1}	θ_2	0.1
kf	{32, 64, 128}	kf	128
d0	{32, 64}	d0_u, d0_b	64, 64
d1	{64, 128}	d1_u, d1_b	64, 64
lr	{0.001, 0.005}	lr	0.005

actv1	{elu, relu, tanh}	actv1	elu
actv2	relu (fixed)	actv2	relu
weight_decay	1e-5 (fixed)	weight_decay	1e-5

Best val RMSE = 1.3606 after Hyperopt search (50/648 combinations ≈ 8% of space explored).

5. Experimental Results

5.1 Experiment Progression

Exp	Key Change	Val RMSE	Test RMSE	Gap
0	Baseline (d0=64, d1=128, kf=64, θ_1=0.001, θ_2=0.001)	1.4189	1.4593	0.33
2	Stronger reg (θ_1=0.1, θ_2=0.01)	1.3656	1.4098	0.31
3	Smaller model (d0=32, d1=64, kf=32, θ_1=0.01, θ_2=0.01)	1.4034	1.4449	0.32
4	Small model + strong reg (θ_1=0.1, θ_2=0.01)	1.3735	1.4162	0.30
Final★	**Hyperopt best (d0=64, d1=64, kf=128, lr=0.005, actv1=elu, θ_1=0.01, θ_2=0.1)**	**1.3606**	**1.4077**	**0.29**
Paper	MG-GAT [FR and NIG interpretable]	—	1.249	—

5.2 Comparison with Paper Baselines

Model	Test RMSE (Paper, PA)	Test RMSE (This Work)
MG-GAT [interpretable]	1.249	1.4077
DGAN	1.250	—
IGMC	1.305	—
GRALS	1.328	—
SVD++	1.339	—

Our result (1.4077) exceeds all paper baselines, but the datasets differ significantly (320K vs 77K users; 2024 vs 2019 Yelp). The gap is primarily attributable to data sparsity — see Section 6.

5.3 User Filtering Ablation

Configuration	Train Reviews	Val RMSE	Test RMSE
Full training data	820,496	1.3606	1.4077
Filtered (≥2 ratings/user)	698,462	1.3729	1.4193

Filtering removed 70.3% of users (only 29.7% had ≥2 training ratings), degrading performance. Full data was used for the final model.

6. Gap Analysis

The gap between our result (1.4077) and the paper (1.249) is approximately 0.159. We decompose it into four factors:

Factor	Est. Contribution	Evidence
Dataset version (sparsity) — primary	~0.08–0.10	70.3% of users have only 1 training rating; NIG ablation reversal (§7) confirms sparsity effect
Missing LLM perceptual map (4th biz graph)	~0.02–0.03	Paper Table 2: removing any graph type → ~0.03 RMSE increase; biz graph avg degree 15.6 vs 30.0
Insufficient HP search	~0.02–0.03	Explored 50/648 combinations (~8%); paper does not disclose search extent
Random seed / initialisation	~0.01–0.02	Stochastic variation
Total estimated	**~0.13–0.18**	**Covers the observed gap of 0.159**

The NIG ablation reversal (§7) provides the strongest evidence for the sparsity hypothesis: in our sparse data, uniform attention outperforms learned attention (Uniform Graph 1.4052 vs full MG-GAT 1.4077), whereas in the paper's denser data, learned weights help significantly (1.249 vs 1.280). This confirms that the attention mechanism requires sufficient per-node observations to learn meaningful weights.

7. Ablation Study

All ablations use the final hyperopt parameters: θ_1=0.01, θ_2=0.1, kf=128, d0=64, d1=64, lr=0.005, actv1=elu.

Configuration	Implementation	Our RMSE	Paper RMSE	Match?
Full MG-GAT	—	1.4077	1.249	—

NIG Removed	$\alpha = 1/\lvert N_i \rvert$ (uniform weights)	1.4093	1.303	✗ Rev.
FR Removed	$H(1) = \text{sigmoid}(W(1) \cdot S)$	1.4087	1.305	✓
Uniform Graph Weighting	$\omega g = 1/3$ fixed	1.4052	1.280	✗ Rev.
No Auxiliary Information	Su, Sb → learnable embeddings	1.4386	1.312	✓
No Networks or Aux. Info	Pure matrix factorisation (H(4) only)	1.4652	1.405	✓

FR Removed: RMSE 1.4077 → 1.4087 (+0.001), consistent with the paper. The linear Layer 1 design incurs a small performance cost for interpretability.

NIG Removed: unlike the paper (+0.054 degradation), our result shows a marginal worsening of +0.0016. With 70.3% single-rating users, the attention mechanism lacks sufficient signal; uniform weighting acts as a stronger regulariser.

Uniform Graph Weighting: fixing $\omega g = 1/3$ yields 1.4052 vs 1.4077 (slightly better in our sparse setting). In the paper's denser data, learned weights help significantly (+0.031). Data sparsity limits the model's ability to learn discriminative graph-level weights.

No Auxiliary Information: RMSE → 1.4386 (+0.031). The paper shows a larger degradation (+0.063), suggesting side features are more valuable in denser data.

No Networks or Auxiliary Information: pure MF achieves worst Test RMSE 1.4652, with severe overfitting (Train ~0.68 vs Test ~1.47) and no early stopping within 200 epochs.

8. Implementation Notes

Design Decision	Paper Basis
Graph regularisation only on H(4), not H(3)	"H(3) is not regularized... because the graph attention design enables local smoothness"
Multi-graph NIG: scores merged before softmax	Eq.3: $\Sigma_g\ \omega_g$ inside LeakyReLU, not separate softmax per graph
3 business graphs stored independently, combined via ωg	"$w_g \in \mathbb{R}$ represents a learnable parameter for each business graph g"
Loss computed only at observed positions	"$\Omega_training \circ (X - \hat{X})$"
SVD uses training data only	Implied by time split; prevents data leakage
Layer 1: no bias, no activation	Linear for FR interpretability — Eq.2
Co-visitation graph from training reviews only	Prevents data leakage
θ_1 = graph reg; $\theta_2 = \tilde{L}$ diagonal; weight_decay = L2	Three separate regularisation roles

9. Conclusion

We reproduced the core MG-GAT architecture (Stage 1) following the paper's specifications. The final model achieves Test RMSE of 1.4077 vs the paper's 1.249. The implementation correctly captures:

- Five-layer interpretable architecture with FR and NIG
- Three business graph types with learnable ωg weights
- Graph Laplacian regularisation applied only to H(4)
- Hyperopt-based hyperparameter search (50 trials × 30 epochs)
- Full ablation study — 6 configurations matching paper Table 2

The ablation study confirms that graph structure and side features both contribute (removing either degrades performance); pure MF is the worst (Test RMSE 1.4652). The NIG and Uniform Graph Weighting ablations show reversed effects vs the paper — the value of learned attention weights depends on having sufficient per-node observations.

The performance gap (~0.159) is primarily attributable to dataset version differences: the 2024 Yelp dataset has 4.2× more users with much higher sparsity (70.3% single-rating users), fundamentally limiting the attention mechanism's ability to learn meaningful neighbor weights — directly evidenced by the NIG ablation reversal. Secondary factors: excluded LLM perceptual map graph and limited HP search coverage (~8% of the space).